

BoraxEVB

This board may have been reworked.
Rework references are in the form "SRWx" and are graphically
denoted by the following symbol.
Reworks are listed and described in detail on page 22.

SRWx



MAN-REWORK

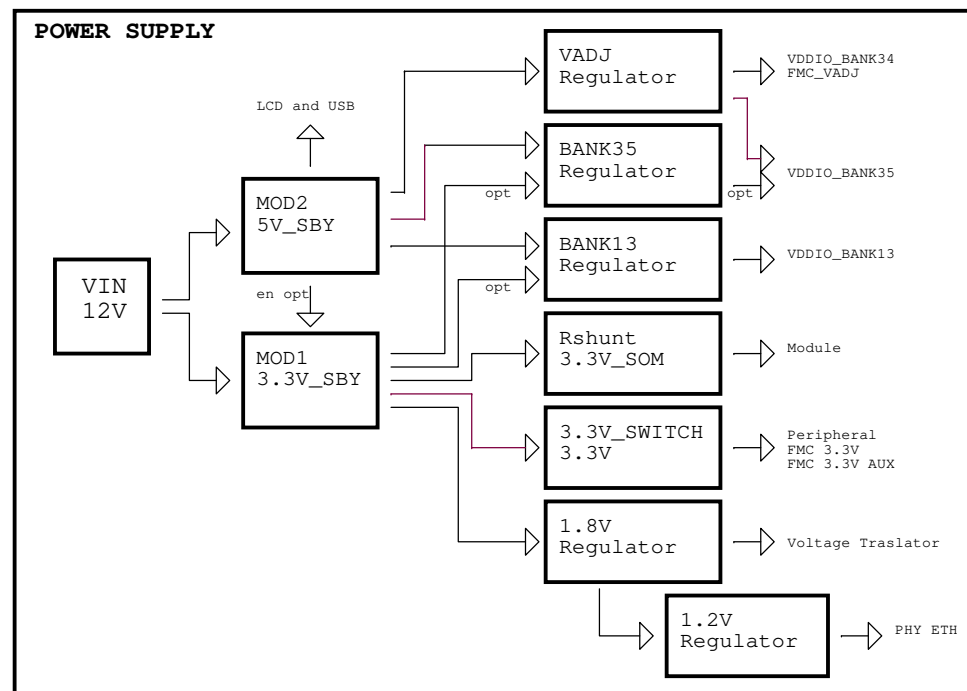
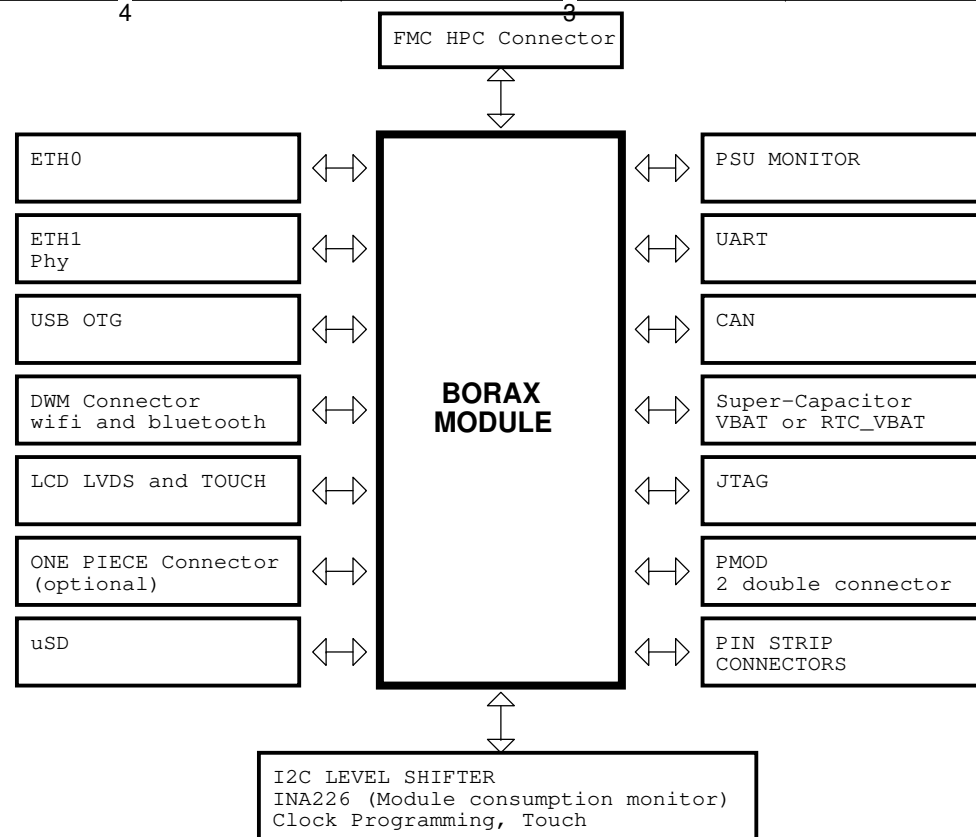
DISCLAIMER

Electrical schematics provided with development kit should be considered as a useful suggestion on how to connect CPU module to peripherals or connectors under a topological point of view. These schemes are provided "AS IS" and shall not be used as they are provided for any reason. Customer must review these schemes to introduce design modifications - starting from the specifications of the system itself - able to achieve stated performance levels.

Gli schemi elettrici forniti con i kit di sviluppo sono da ritenersi un utile riferimento per quanto riguarda la topologia di collegamento elettrico tra il modulo CPU, i connettori e le periferiche ad esso collegati. Tali schemi sono forniti "AS IS" e non devono essere ritenuti per nessun motivo impiegabili così come essi sono forniti. Rimane di esclusiva competenza del cliente la responsabilità di introdurre a tutti i livelli gli accorgimenti progettuali che - a partire dalle specifiche di progetto del sistema stesso - consentono il raggiungimento di determinati livelli di prestazione.

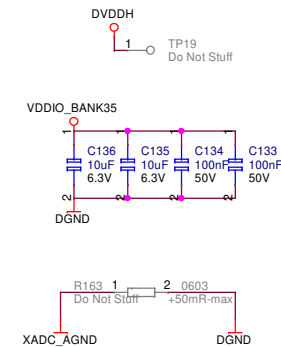
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		DAVE Embedded Systems via Talponedo 29/A 33080 Porcia (PN) - Italy www.dave.eu	
Title BoraxEVB		PCB Code CS143714	
Size A4	Document Number Cover	Sch Rev See ->	Page Rev 1.6.1
Date: Wednesday, June 12, 2019		Sheet 1	of 23



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		DAVE Embedded Systems via Talponedo 29/A 33080 Porcia (PN) - Italy www.dave.eu	
Title BoraxEVB		PCB Code See COVER	
Size A3	Document Number Block diagram	Sch Rev See COVER	Page Rev 1.0.2
Date: Wednesday, December 14, 2016		Sheet 2	of 23

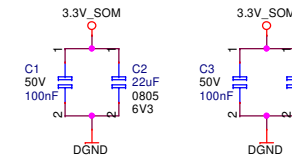
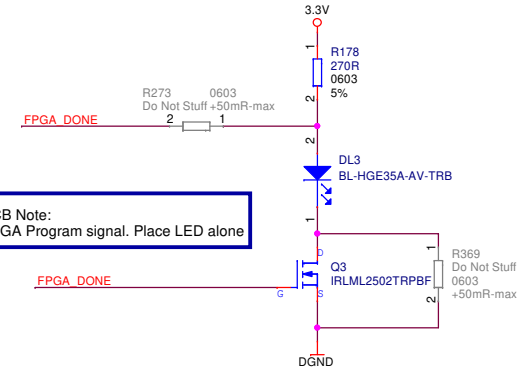
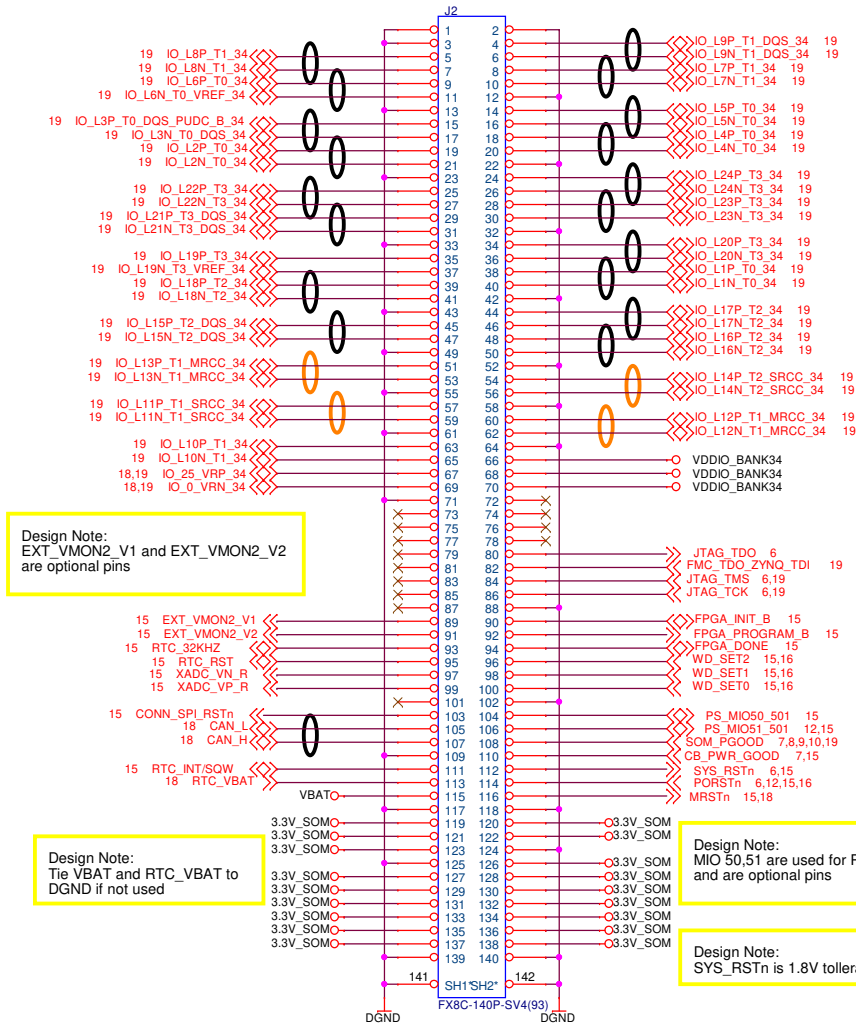


BANK35 is HP I/O type for 7030 device
Min voltage 1.2V
Max voltage 1.8V

I2C LEVEL SHIFTER

Design Note:
See also Xilinx AR# 61861
and I2C glitch filter workaround on DAVE Wiki:
http://wiki.dave.eu/index.php/Integration_guide_%28Bora%29PS.27_I.C2.B2C_buses

 DAVE Embedded Systems		DAVE Embedded Systems via Talponedo 29/A 33080 Porcia (PN) - Italy www.dave.eu	
Title BoraxEVB		PCB Code See COVER	
Size A3	Document Number Borax Connector 1/3	Sch Rev See COVER	Page Rev 1.6.1
Date Wednesday, June 12, 2019		Sheet 3	of 23

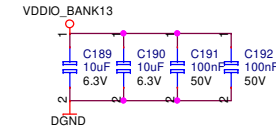


DOCUMENTATION NOTE:
BANK34 is HR I/O type for 7015 device
Min voltage 1.2V
Max voltage 3.3V

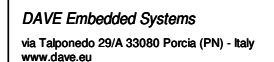
BANK34 is HP I/O type for 7030 device
Min voltage 1.2V
Max voltage 1.8V

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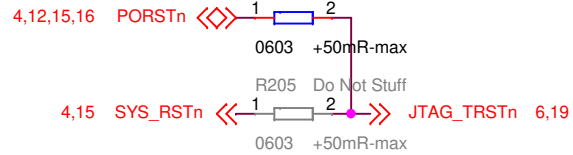
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Title	BoraxEVB	PCB Code See COVER	
Size	A3	Document Number Borax Connector 2/3	Sch Rev See COVER
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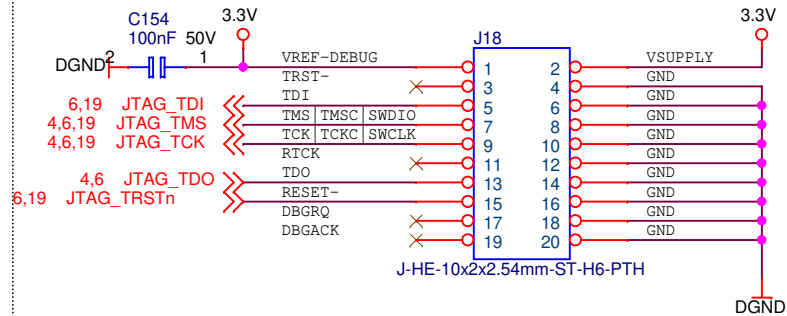
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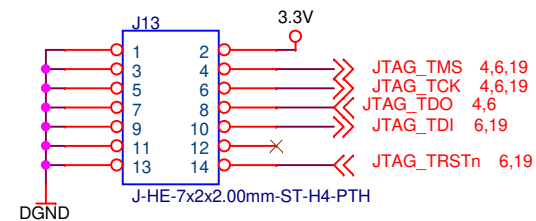
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BoraxEVB		See COVER	
Size	Document Number	Sch Rev	Page Rev
A3	Borax Connector 3/3	See COVER	1.0.4
Date:	Wednesday, December 14, 2016	Sheet	5 of 23



JTAG (ARM 20-pin connector)



JTAG (XILINX 14-pin connector)



PCB Notes:
For OUTLINE see XILINX Jtag Connector

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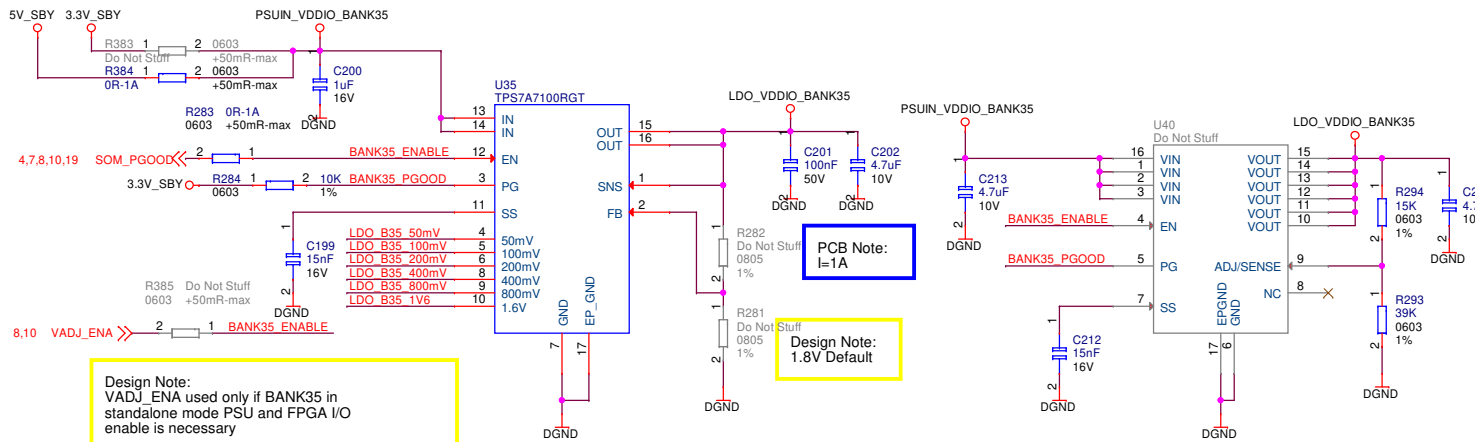


DAVE Embedded Systems

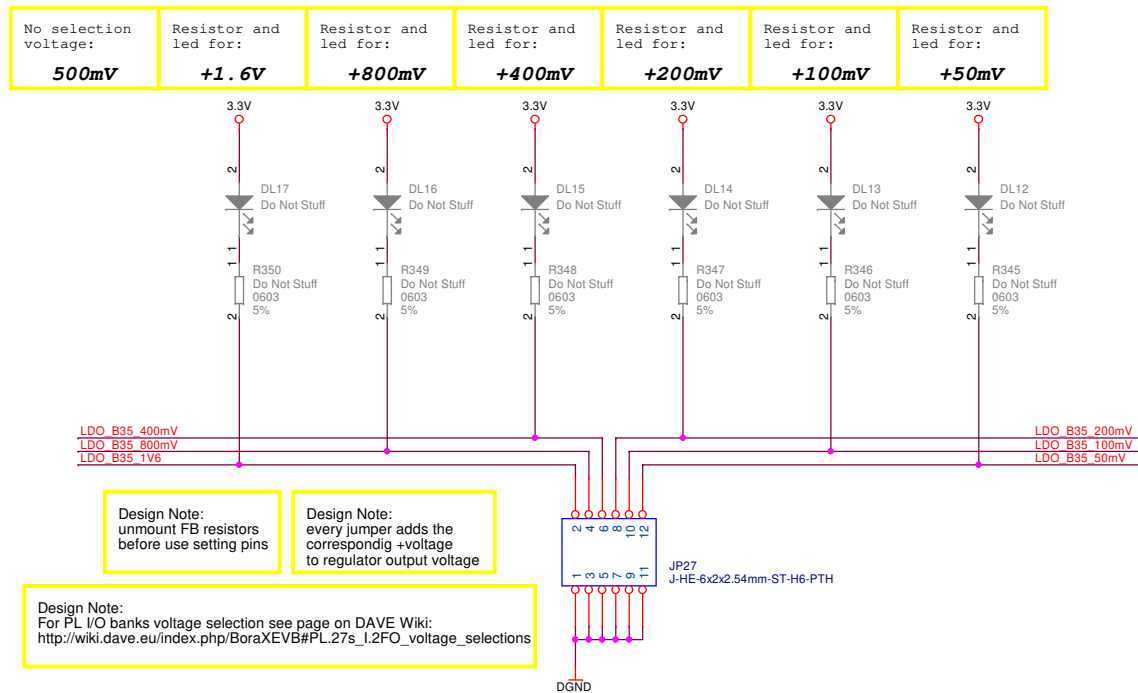
via Talponedo 29/A 33080 Porcia (PN) - Italy
www.dave.eu

Title BoraxEVB		PCB Code See COVER	
Size A4	Document Number JTAG	Sch Rev See COVER	Page Rev 1.0.0
Date:	Wednesday, December 14, 2016	Sheet	6 of 23

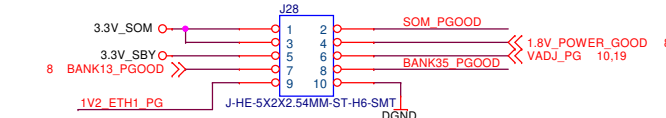
LDO VDDIO BANK35 Regulators



LDO BANK35 REGULATOR OUTPUT SETTINGS



BANKS PGOODs

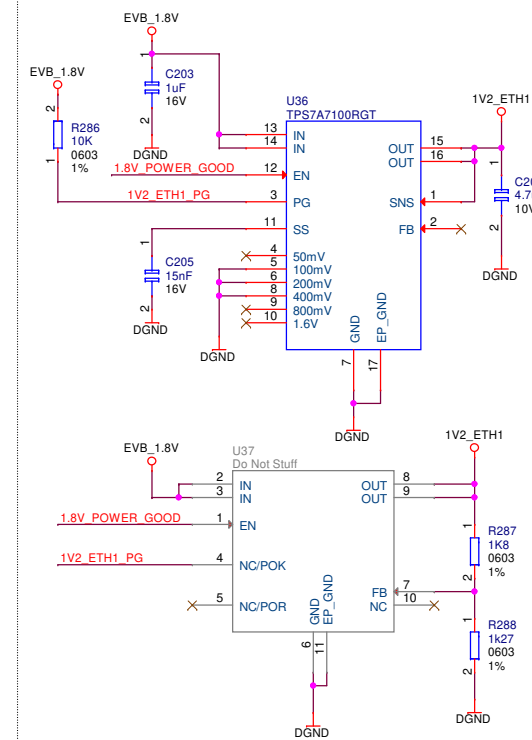


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DAVE Embedded Systems
via Talponedo 29/A 33080 Porcia (PN) - Italy
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Title BoraxEVB		PCB Code See COVER	
Size A3	Document Number PSU 2/2	Sch Rev See COVER	Page Rev 1.0.4
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1.2V Ethernet Core Supply



U43 Do Not Stuff

VADJ pin configuration

PCB Notes:

- Connect C25 DGND pin to EP - I=2A

Components and Connections:

- VADJ_PG:** 9.19, SOM_PG00D, 5V_SBY, R310 (100K 0603 1%), R357 (0R-1A 0603 +50mR-max).
- VADJ_ENA:** 4.7, 8.9, 19, 5V_SBY, R309 (0603 0R-1A +50mR-max), R358 (0603 Do Not Stuff +50mR-max).
- VADJ_FB:** R308 (0603 1% Do Not Stuff), C221 (22pF), R312 (100K 0603 1%).
- VADJ_ENA:** 8.9, R358 (0603 Do Not Stuff +50mR-max).
- Other Components:** C220 (10uF 6.3V), C25 (22pF), C26 (22pF), C27 (22pF), C28 (22pF), C29 (22pF), C30 (22pF), C31 (22pF), C32 (22pF), C33 (22pF), C34 (22pF), C35 (22pF), C36 (22pF), C37 (22pF), C38 (22pF), C39 (22pF), C40 (22pF), C41 (22pF), C42 (22pF), C43 (22pF), C44 (22pF), C45 (22pF), C46 (22pF), C47 (22pF), C48 (22pF), C49 (22pF), C50 (22pF), C51 (22pF), C52 (22pF), C53 (22pF), C54 (22pF), C55 (22pF), C56 (22pF), C57 (22pF), C58 (22pF), C59 (22pF), C60 (22pF), C61 (22pF), C62 (22pF), C63 (22pF), C64 (22pF), C65 (22pF), C66 (22pF), C67 (22pF), C68 (22pF), C69 (22pF), C70 (22pF), C71 (22pF), C72 (22pF), C73 (22pF), C74 (22pF), C75 (22pF), C76 (22pF), C77 (22pF), C78 (22pF), C79 (22pF), C80 (22pF), C81 (22pF), C82 (22pF), C83 (22pF), C84 (22pF), C85 (22pF), C86 (22pF), C87 (22pF), C88 (22pF), C89 (22pF), C90 (22pF), C91 (22pF), C92 (22pF), C93 (22pF), C94 (22pF), C95 (22pF), C96 (22pF), C97 (22pF), C98 (22pF), C99 (22pF), C100 (22pF).

```

graph TD
    VADJ[VADJ] -- OR --> VDDIO_BANK34[VDDIO_BANK34]
    VDDIO_BANK34 -- OR --> FMC_VADJ[FMC_VADJ]
    VDDIO_BANK34 -- OR --> VDDIO_BANK35[VDDIO_BANK35]
    VDDIO_BANK35 -- OR --> VDDIO_BANK13[VDDIO_BANK13]
    LDO[LDO] -- OR --> VDDIO_BANK35
    DC_DC[DC/DC] -- OR --> VADJ

```

Resistor and led for selection:
3.3V

Resistor and led for selection:
2.5V

Resistor and led for selection:
1.8V

Resistor and led for selection:
1.5V

Resistor and led for selection:
1.2V

Resistor and led for selection:
RFU

Design Note:
For PL I/O banks voltage selection see page on DAVE Wiki:
http://wiki.dave.eu/index.php/BoraXEVb#PL_27s_I2FO_voltage_selections

JP28
J-HE-6x2x2.54mm-ST-H6-PTH

DGND

LDO_VDDIO_BANK35 VDDIO_BANK35

R344 Do Not Stuff

1 2

1206 +50mR-max

3.3V

FMC_3P3V

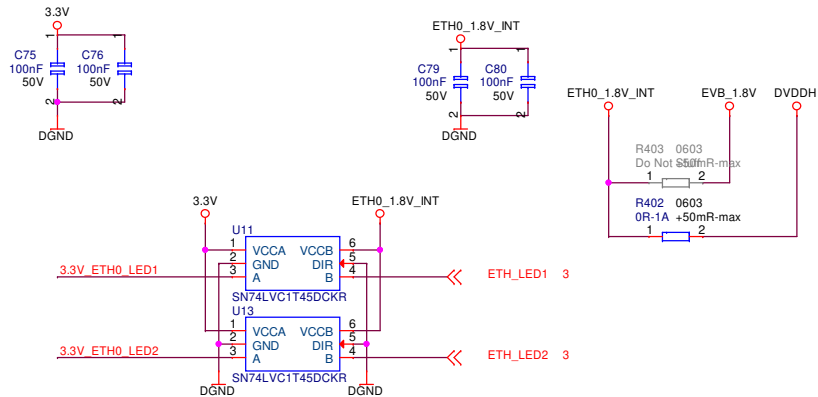
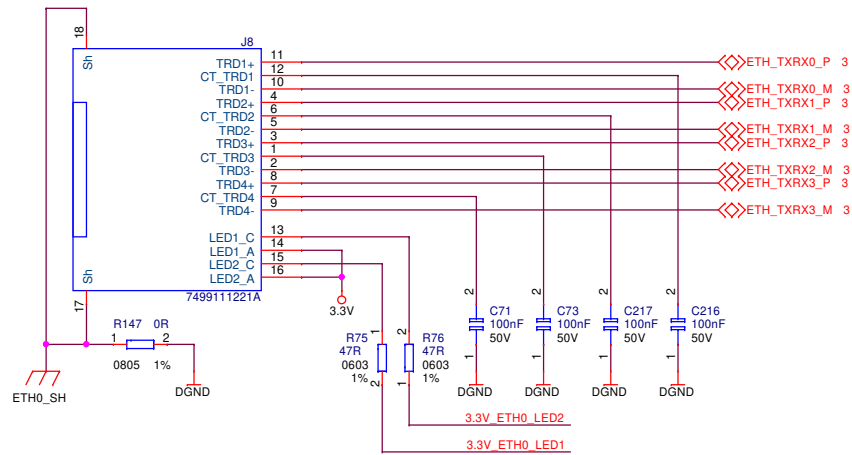
R305 0R-2A

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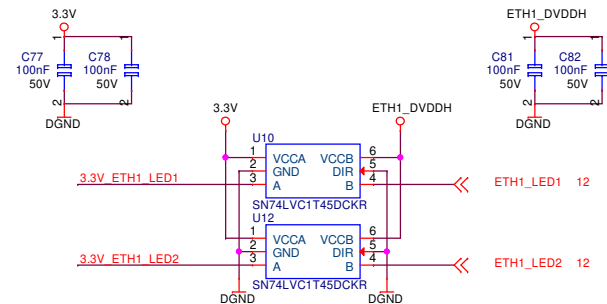
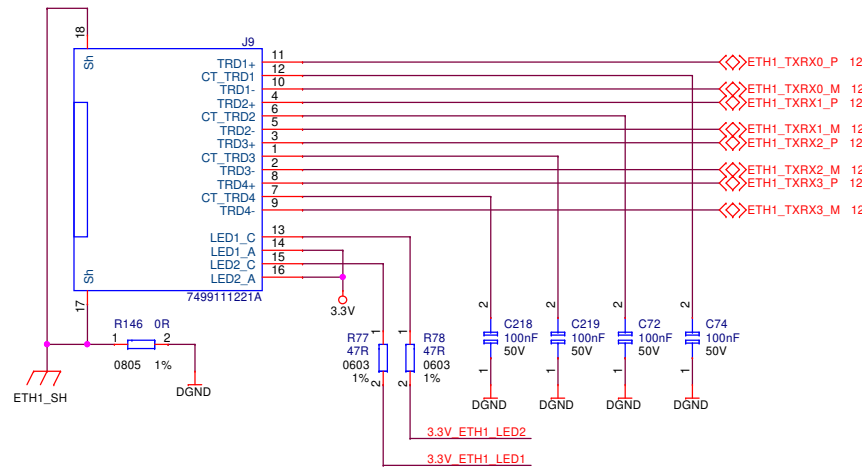
1206 +50mR-max

PCB Note:
I=3A max

ETH0 Connector



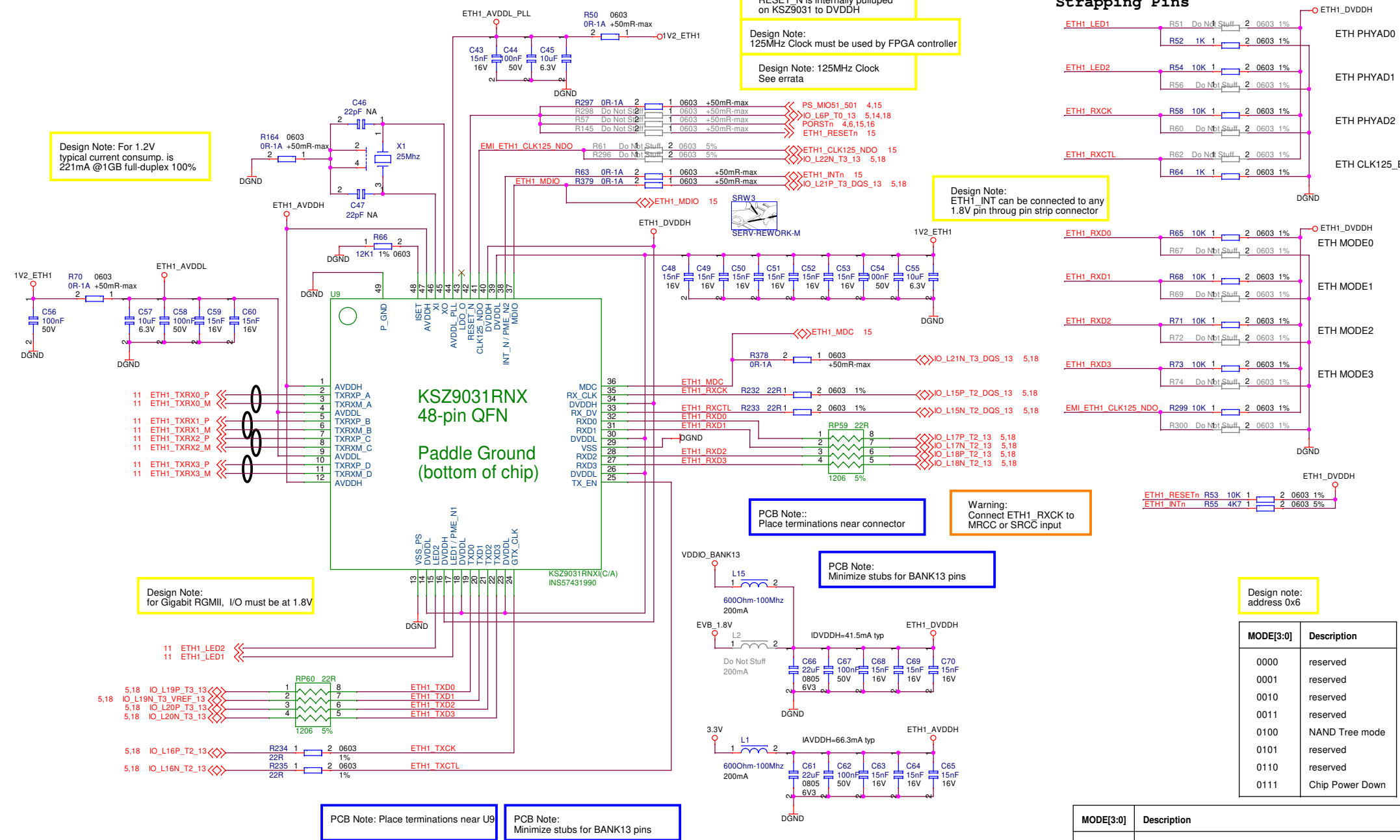
ETH1 Connector



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		DAVE Embedded Systems via Talponedo 29/A 33080 Porcia (PN) - Italy www.dave.eu	
Title	BoraxEVB	PCB Code	See COVER
Size	A3	Document Number	See COVER
Date:	Wednesday, December 14, 2016	Sch Rev	Page Rev
		Ethernet Connectors	1.0.0
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ETHERNET PHY 1GB

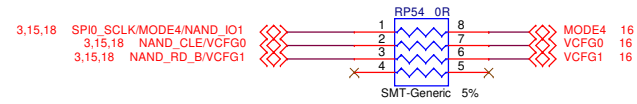
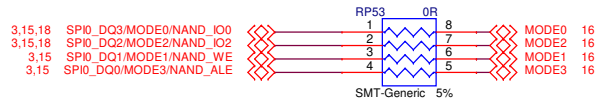


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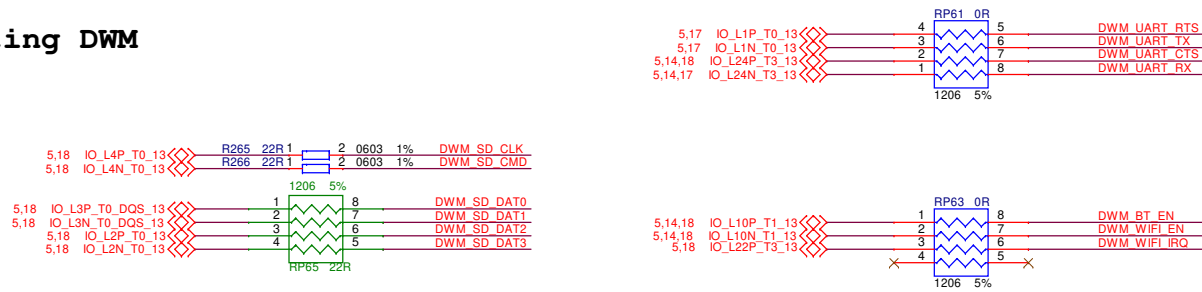
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Title		PCB Code	
BoraxEVB		See COVER	
Size	Document Number	Sch Rev	Page Rev
A3	Ethernet PHY EVB #1	See COVER	1.6.0
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MODE[3:0]	Description
1000	reserved
1001	reserved
1010	reserved
1011	reserved
1100	Advertise 1000BT full-duplex only (RGMII)
1101	Advertise 1000BT full- and half-duplex only (RGMII)
1110	Advertise all capabilities, except 1000BT half-duplex (RGMII)
1111	Advertise all capabilities (RGMII)

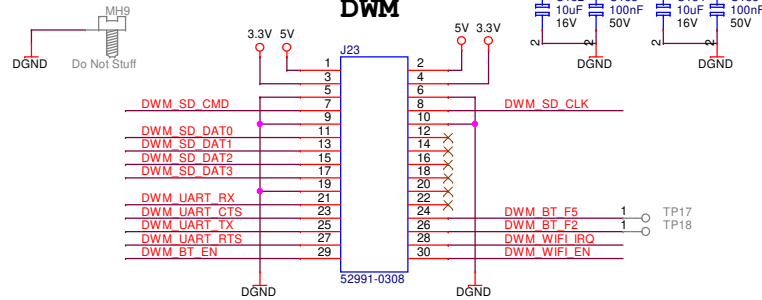
BOOTSTRAP



Routing DWM



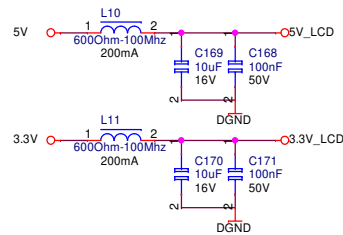
Mounting Hole 2.7mm



PCB Note: Controlled Impedance
50ohm on SDIO lines

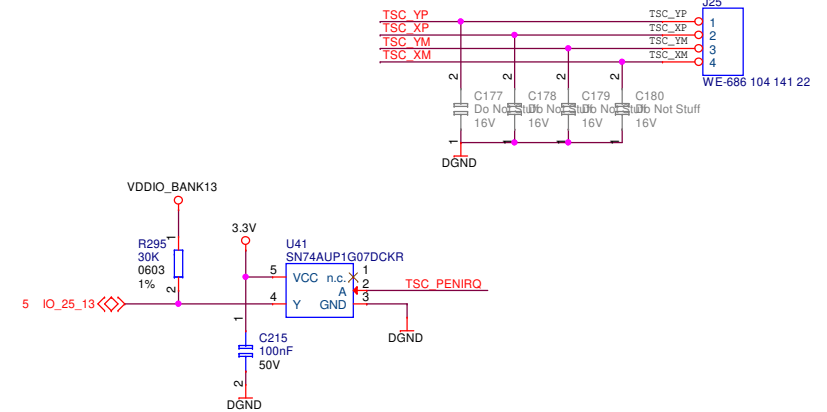
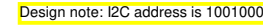
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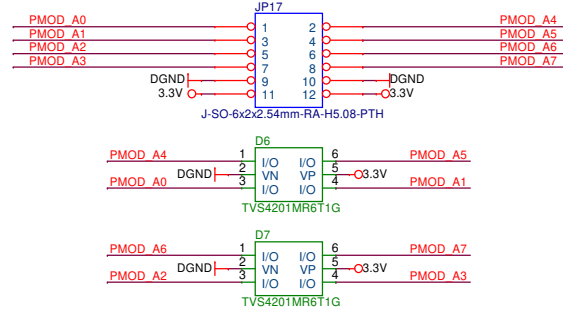


```
# 0R -> LCD Ampire AM-800480STMQW-TA0
## 0R -> LCD Ampire AM-800480STMQW-TA0
### 0R -> LCD Ampire AM-800480R3TMQW-TA1H-A
```

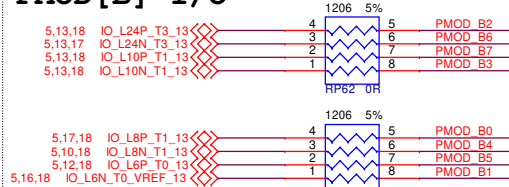




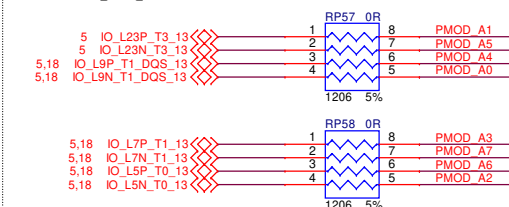
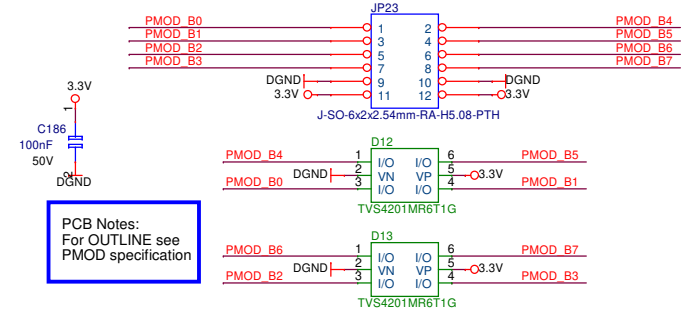
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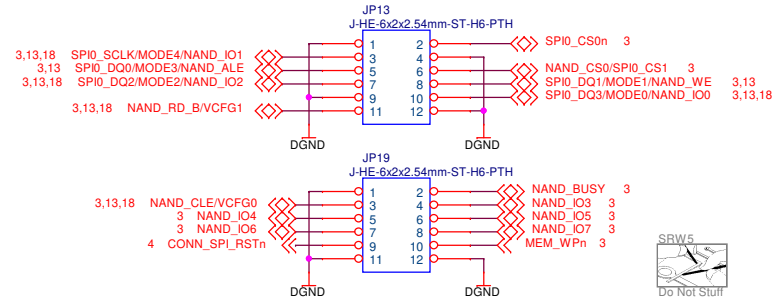


Title		PCB Code	
BoraxEVB		See COVER	
Size	Document Number	Sch Rev	Page Rev
A3	Routing PMOD, LCD, Touch	See COVER	1.0.4
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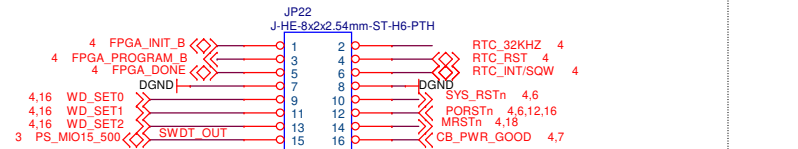
PIN STRIP CONNECTORS

PCB Notes:
Place all Pin Strip at multiple 2.54mm spacing

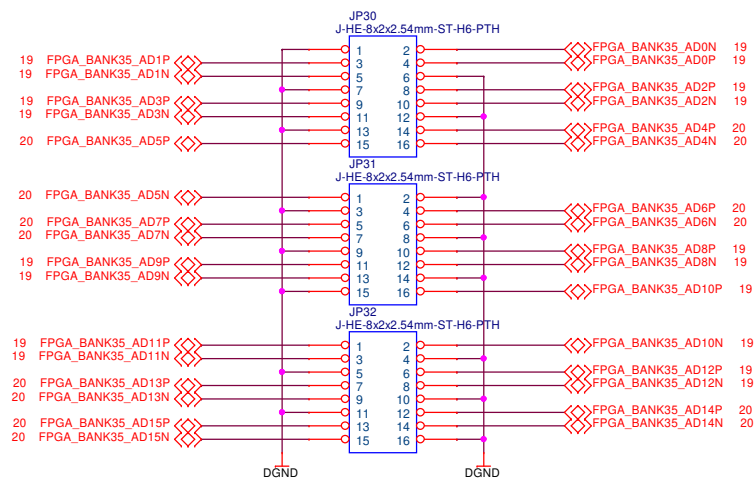
BANK 500 (SPI, NAND)



FPGA, WD, RTC, RST

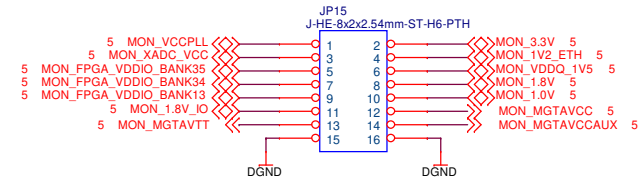


ADC FROM FPGA BANK35

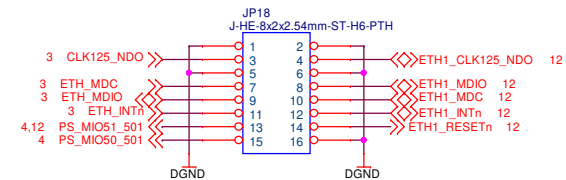


PCB Notes:
Place all Pin Strip at multiple 2.54mm spacing

VOLTAGE MONITOR

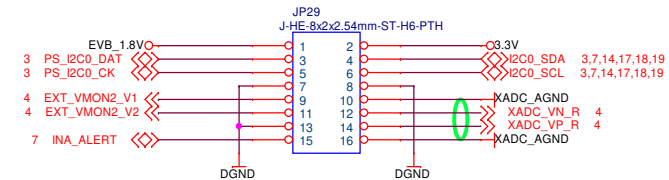


ETHERNET GPIO



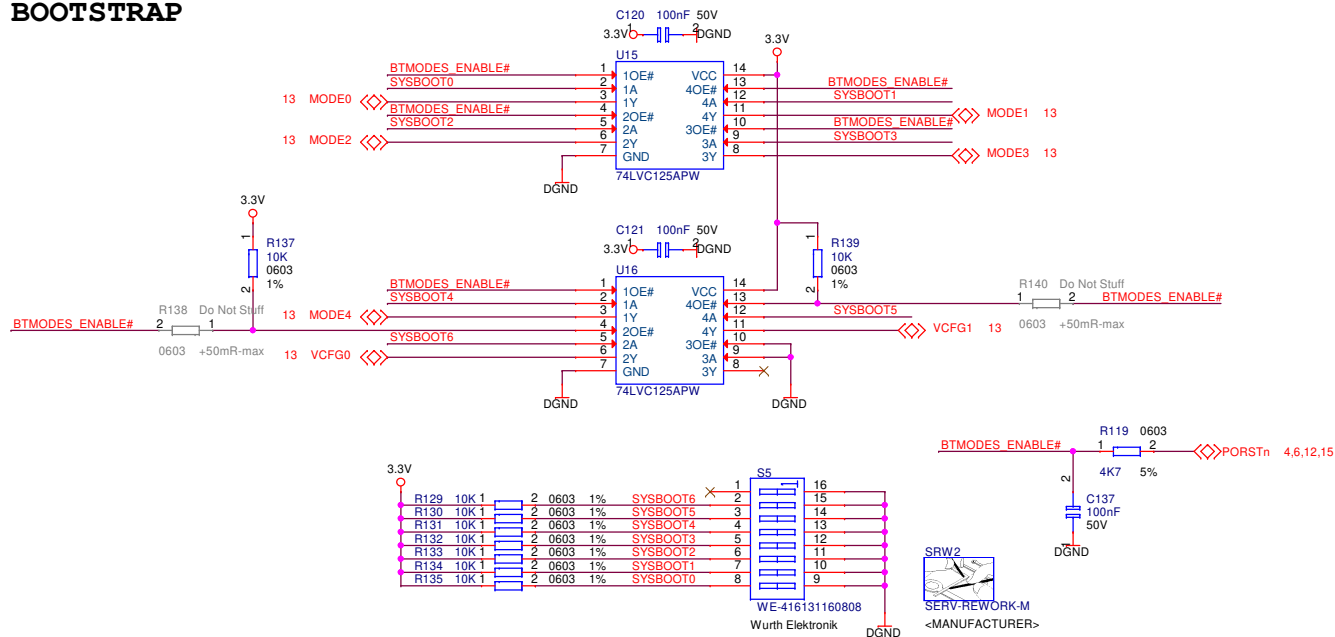
AUX PINS

I2C, Aux BORAX Voltage Monitor Pins, INA Alert, XADC

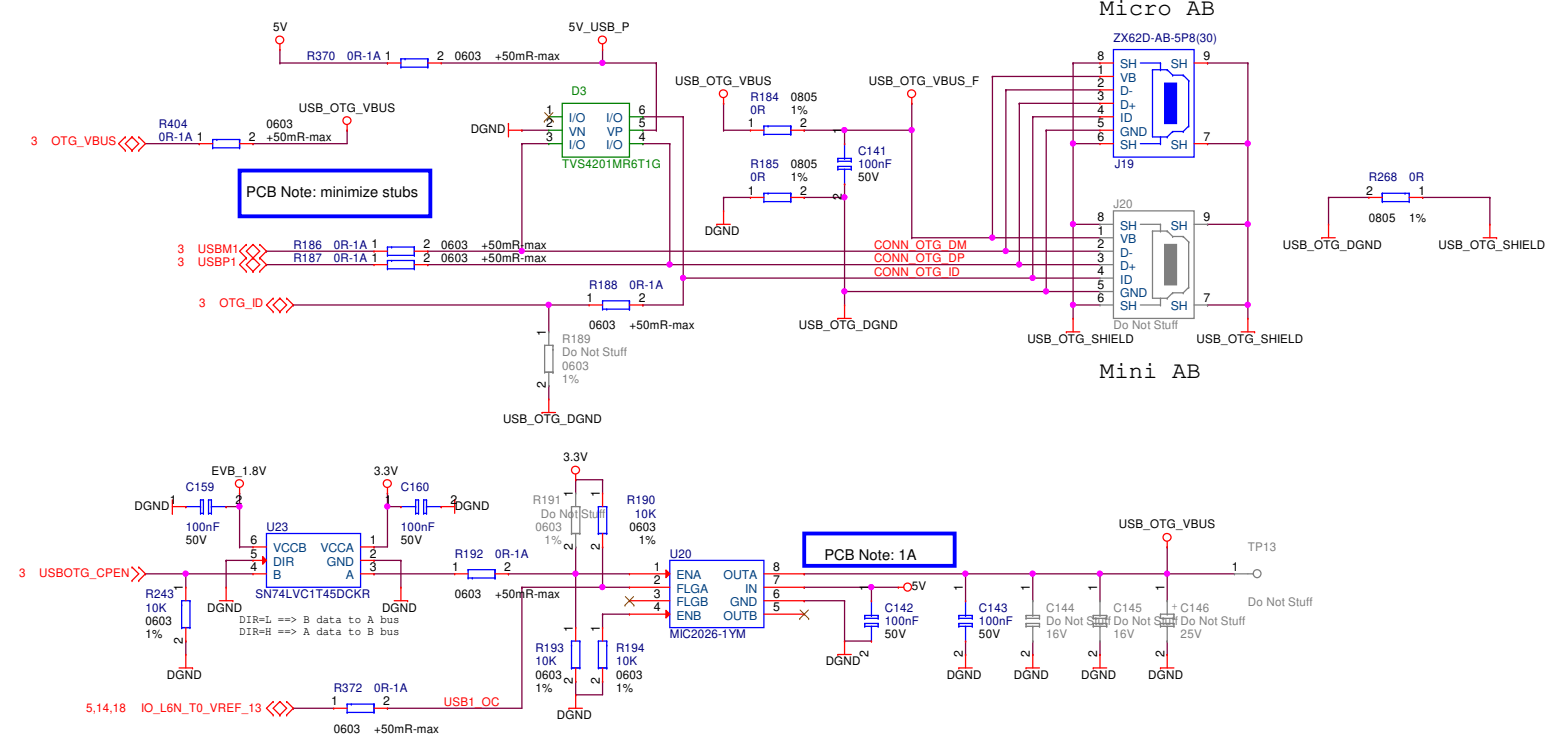


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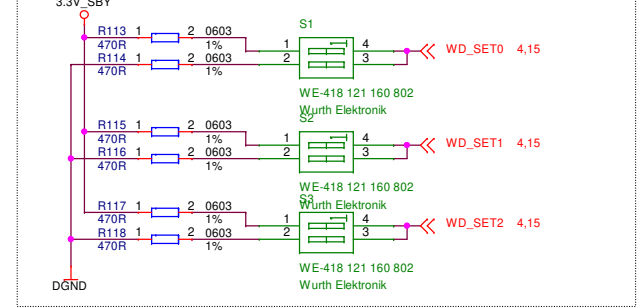
BOOTSTRAP



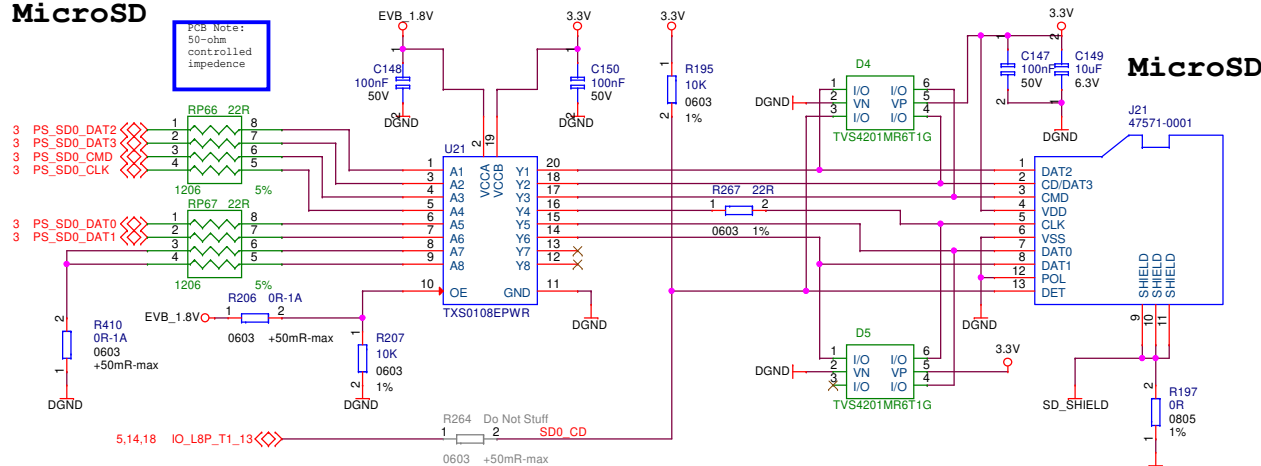
USB OTG



WD SETTINGS



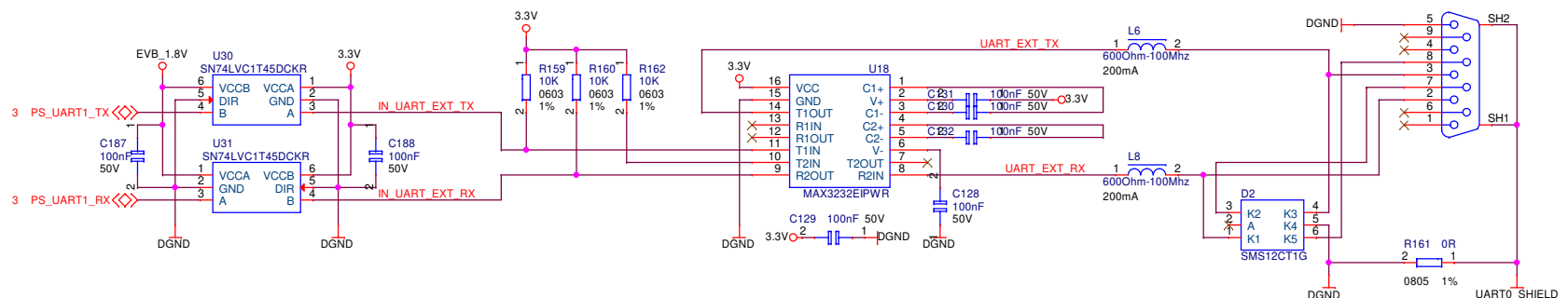
MicroSD



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File	BoraxEVB	PCB Code	See COVER
Size	A3	Document Number	Sch Rev
		UART, uSD, prog.clock	Page Rev
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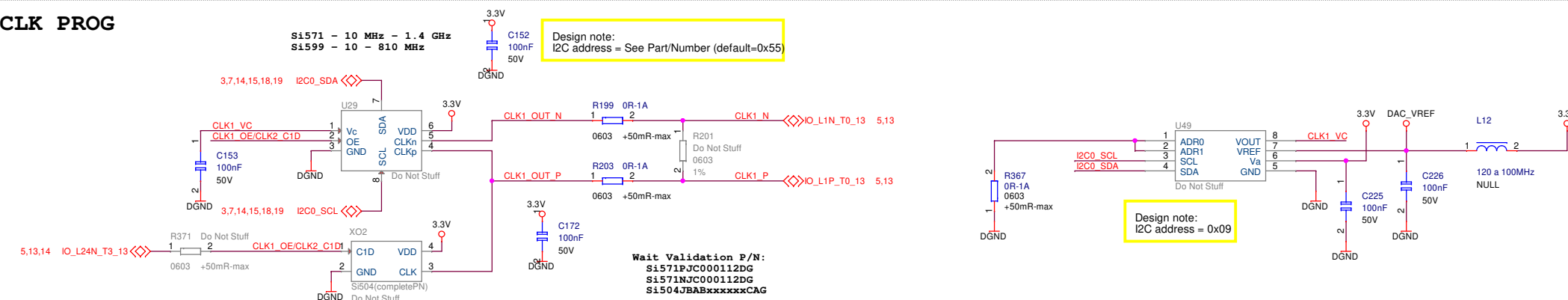
UART1 (console)



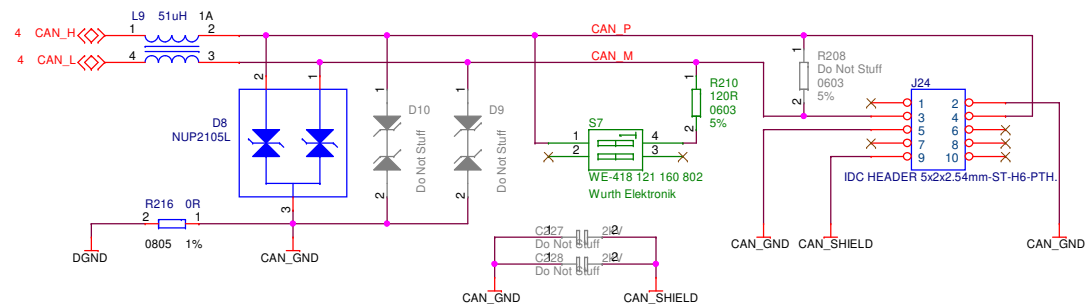
CLK PROG

Si571 - 10 MHz - 1.4 GHz
Si599 - 10 - 810 MHz

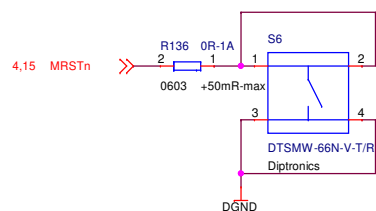
Design note:
I2C address = See Part/Number (default=0x55)



CAN



RESET



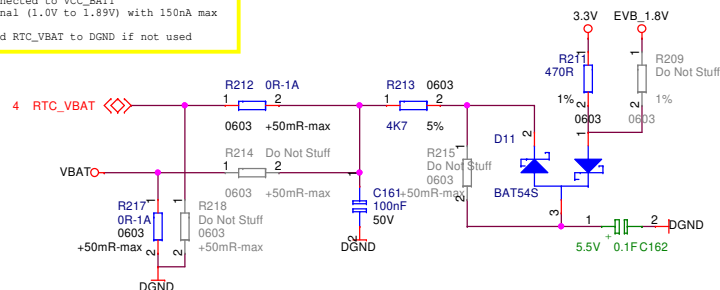
Super Cap

```

RTC_VBAT is connected to DS3232M
3.0V nominal (2.3V to 4.5V)
VBAT is connected to VCC_BATT
1.5V nominal (1.0V to 1.89V) with 150nA max

Tie VBAT and RTC_VBAT to DGND if not used

```



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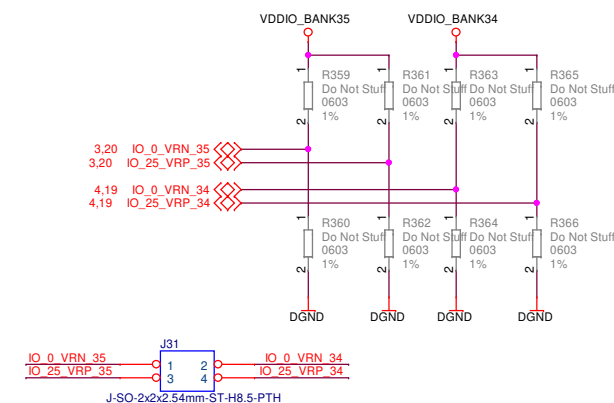


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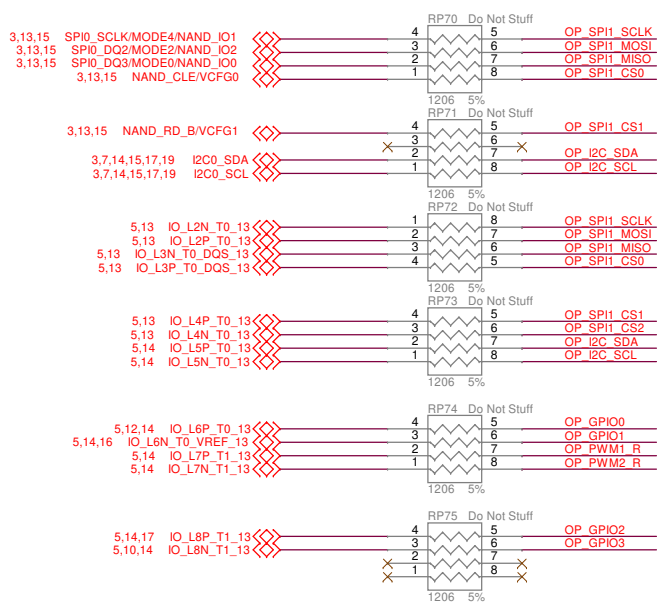
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BANK 34, 35 DCI reference pins

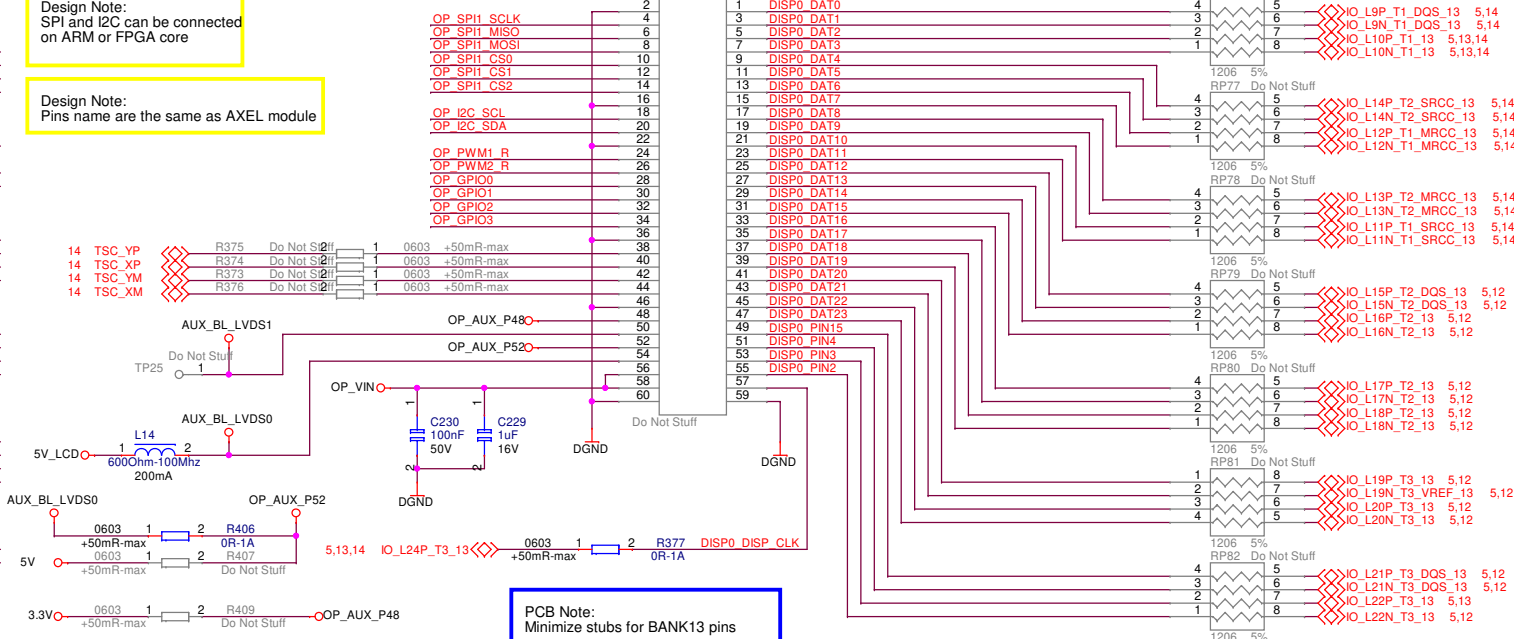


ONE PIECE CONNECTOR



Design Note:
SPI and I2C can be connected
on ARM or FPGA core

Design Note:
Pins name are the same as AXEL module



PCB Note:
Minimize stubs for BANK13 pins

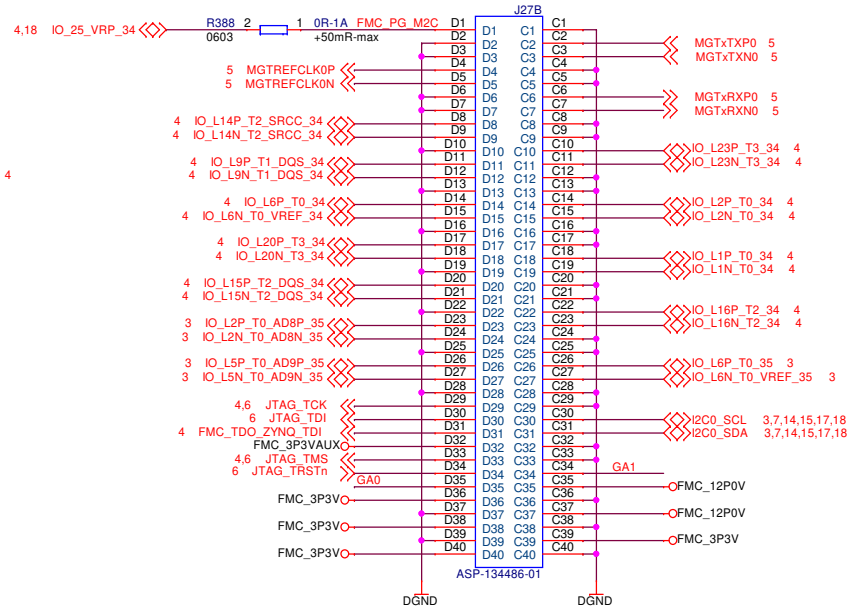
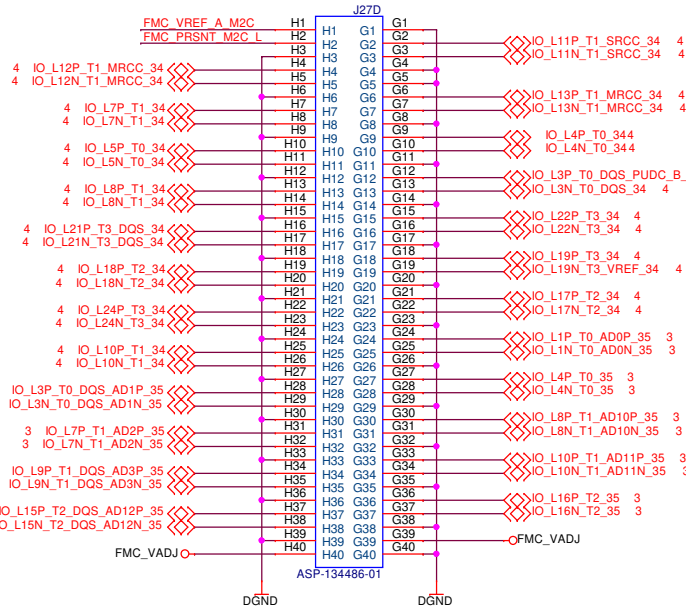
PCB Note:
symbols can be found in
<http://www.samtec.com/standards/vita.aspx>

Design Note:
SRCC and MRCC pins are NOT SWAPPABLE

Design Note:
- LPC compliance: full
- HPC compliance: partial
- Gigabit: 4x

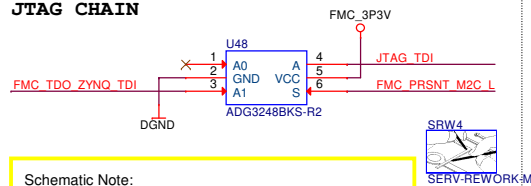
Design Note:
From VITA SPECS Rule 5.35:
Only gigabit data pins with an ordinal
value of '0' are available on the low-pin
count connector.

Design Note:
Clocks pin's direction may change
between version 1.0 and 1.1

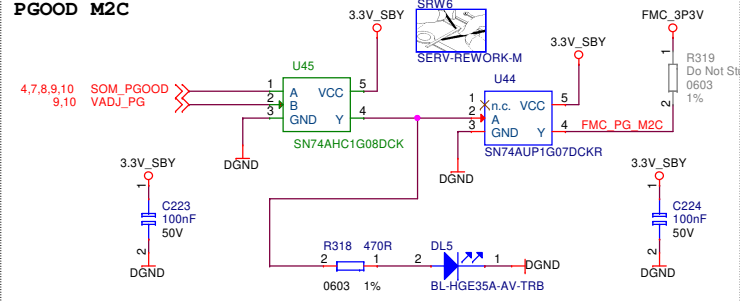


Design Note:
FMC_12P0V power rail level, when
using 12.00V or below input source
and DIODE protection,
is under 5% voltage.

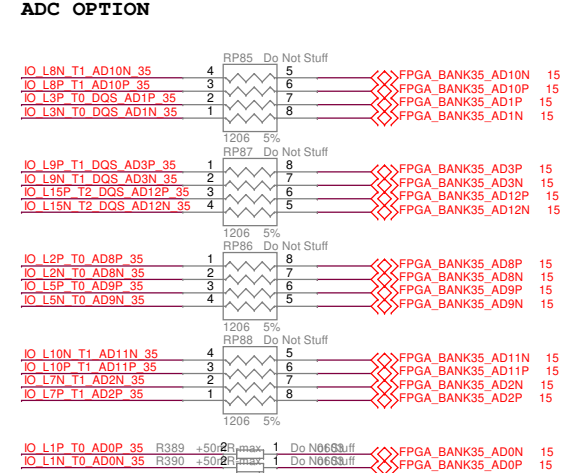
JTAG CHAIN



PGOOD M2C



ADC OPTION

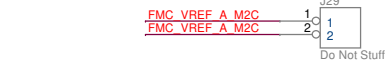


PCB Note:
Minimize stubs on FPGA side

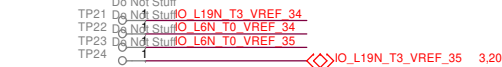
MEZZANINE PRSNT SIGNAL



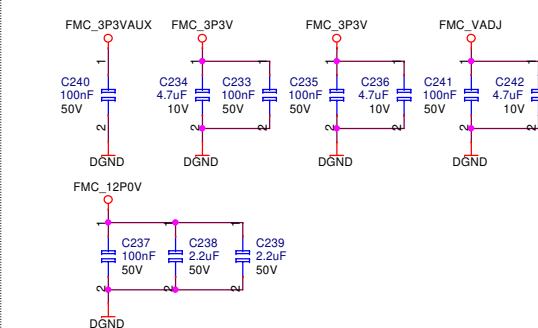
FMC VREF FROM MEZZANINE CARD



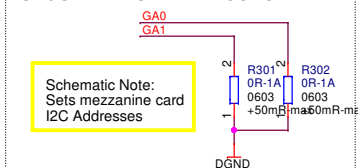
ZYNQ VREF TP PREDISPOSITION



FMC BYPASS

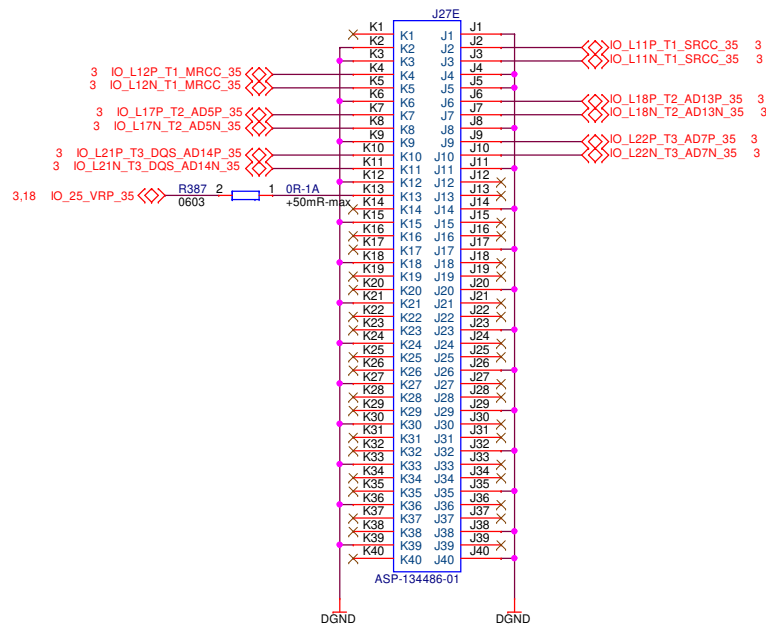


GEOGRAPHIC ADDRESSES

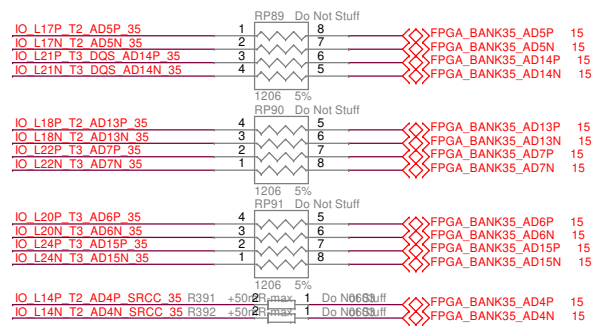


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Design Note:
Follows VITA Rule 5.15: First populate LA, then HA.



ADC OPTION

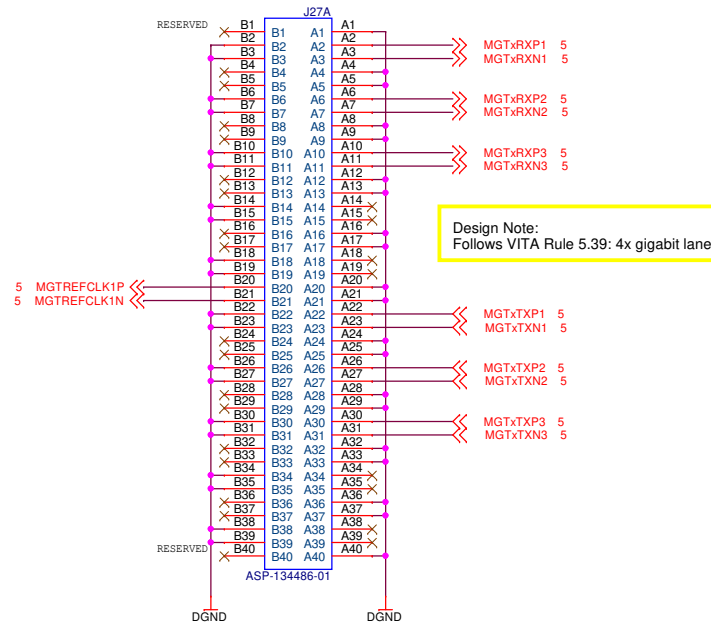
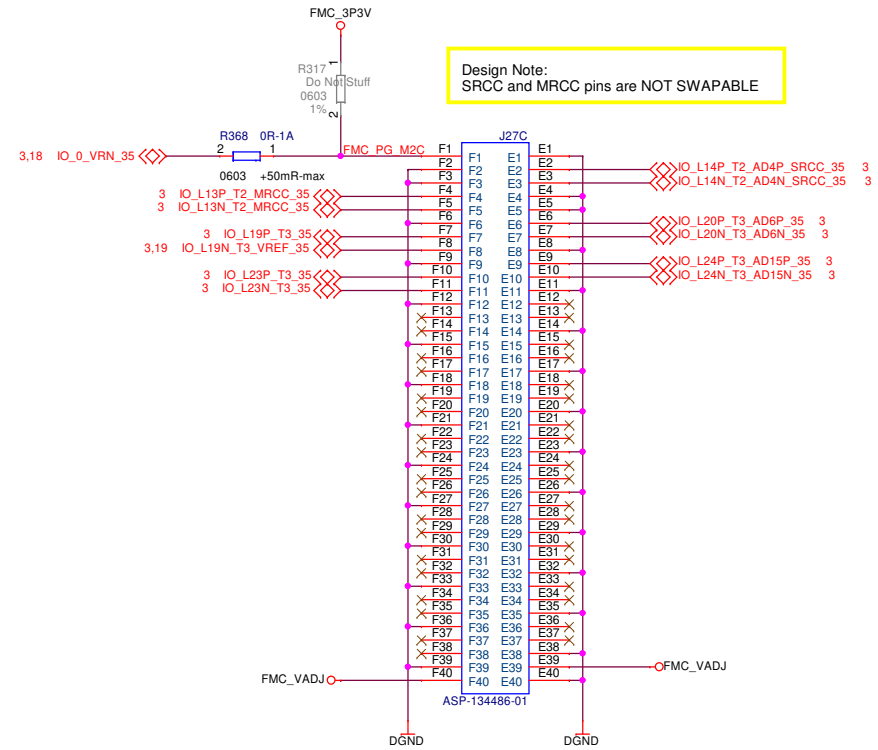


Design Note:

- LPC compliance: full
- HPC compliance: partial
- Gigabit: 4x

PCB Note:
symbols can be found in
<http://www.samtec.com/standards/vita.aspx>

Design Note:
SRCC and MRCC pins are NOT SWAPABLE



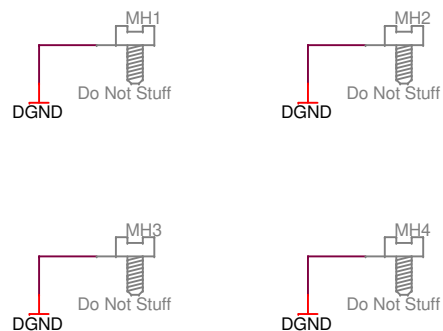
S.EVBBX0000C0R.1.6.0



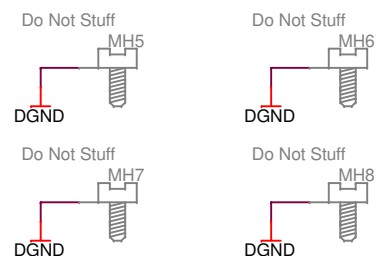
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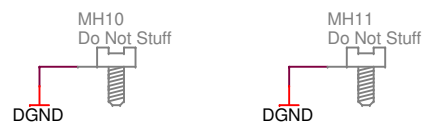
Holes for SOM



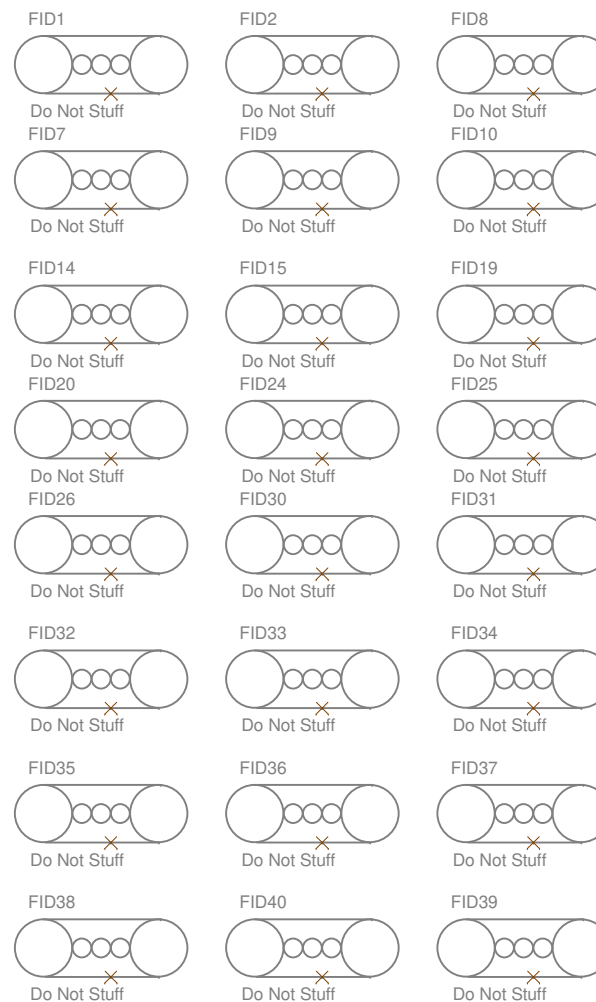
Holes for EVB



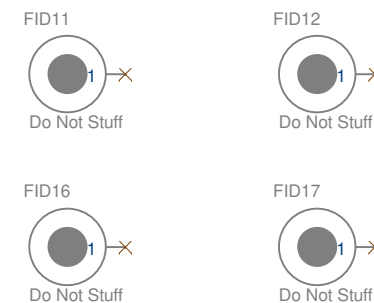
Holes for FMC



Breakaway for figures



Fiducial for module



PCB



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Reference	Description	Device description / additional notes	DAVE code (if any)
SRW1	DipSwitch S1,S2,S3 config.: S1.1 ON S3.2 ON others OFF		
SRW2	DipSwitch S5 configuration: S5.1 OFF S5.2 ON S5.3 OFF S5.4 ON S5.5 OFF S5.6 ON S5.7 ON S5.8 OFF		
SRW3	Connect 1K resistor from R379.2 to C66.1		RAB021R
SRW4	Connect 10K resistor from U48.6 to U48.5		RAA003R
SRW5	20K resistor from JP19.6 to JP19.12 20K resistor from JP19.7 to JP19.12 20K resistor from JP19.2 to JP19.12 20K resistor from JP19.4 to 3.3V (R137.1) 20K resistor from JP19.8 to 3.3V (R137.1)		RAB215R
SRW6	10K resistor from R319.2 to R342.2		RAB000R

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Known issues

1) LCD_BKLT_PWM signal is derived from IO_0_13 on BANK13.
In the case of LVDS signals for LCD this BANK must be powered at 2.5V.
So in this case LCD_BKLT_PWM is an LVCMOS 2.5V signal.
Place a voltage level translator to 3.3V if the signal voltage levels are not compatible
with the LCD diplay backlight input.

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